

Zhen Chen

Seattle, WA

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Research Summary

Researcher specializing in computer graphics, geometry processing, physically-based simulation, 3D generative systems, and embodied AI. My work focuses on modeling complex physical phenomena and building scalable, physically grounded systems for content creation, synthetic 3D data generation, interactive applications, and controllable generative workflows. I have experience bridging research and production, and am interested in applying these techniques to generative AI, spatial intelligence, video/world models, embodied AI, robotics, and agentic content creation systems.

Education

University of Texas at Austin

Ph.D. in Computer Science

Advisor: Etienne Vouga

Austin, TX

2018 – 2024

University of Science and Technology of China

B.S. in Information and Computing Science

Hefei, China

2014 – 2018

Employment

Adobe Research

Research Scientist/Engineer II

Seattle, WA

2024 – Present

- **3D Synthetic Data & Physical AI:** Developing scalable 3D data physically grounded generation workflows for synthetic scenes, camera/view control, dynamic scene construction, rendering automation, metadata export, and model-facing data pipelines for generative, spatial, and embodied AI systems.
- **Animation Systems (Adobe Express):** Led the development and launch of real-time physics-driven animation systems, enabling scalable and interactive motion effects used by millions of users. Designed systems balancing physical realism, performance, and responsiveness for real-time applications.
- **Controllable Generative Simulation:** Developed simulation systems integrated with learning-based and language-guided components for controllable generation of physical behaviors, enabling agentic workflows where high-level user intents (e.g., motion style or “squishiness”) are mapped to physically consistent parameters.
- **Simulation Systems:** Designed and deployed high-performance simulation systems for deformable objects, with a focus on robustness, numerical stability, and scalability in production environments, with applications to physically realistic interaction modeling.
- **Generative Pipeline:** Developed pipelines combining simulation and generative models for scalable 3D content creation and asset generation.

Tencent America

Research Intern

Bellevue, WA

Summers 2022, 2023

- Developed robust low-poly mesh generation techniques for general 3D models, handling challenging cases such as small gaps, holes, and non-manifold geometry.

- Built reliable geometry processing pipelines for real-world assets, emphasizing robustness, efficiency, and scalability, with applications to large-scale and interactive environments.

Adobe
Research Intern

Remote / Austin, TX
Summer 2021

- Developed stable time integration schemes for physically-based simulation under challenging contact scenarios.

Publications

3D Generation & Learning-Based Methods

- *ReWeaver: Simulation-Ready and Topology-Accurate Garment Reconstruction.*
Ming Li, Hui Shan, ..., **Zhen Chen**, Xiangru Huang. CVPR, 2026.
- *Interact3D: Compositional 3D Generation of Interactive Objects.*
Hui Shan, Keyang Luo, ..., **Zhen Chen**, Xiangru Huang. arXiv, 2026.
- *Neural Volumetric Mesh Generator.*
Yan Zheng, Lemeng Wu, ..., **Zhen Chen**, Qixing Huang. NeurIPS Workshop, 2022.

Geometry Processing

- *Robust Low-Poly Meshing for General 3D Models.*
Zhen Chen, Zherong Pan, Kui Wu, Etienne Vouga, Xifeng Gao. ACM TOG (SIGGRAPH), 2023.
- *MiNT: Discretely Integrable Moments for Symmetric Frame Fields.*
Josh Vekhter, **Zhen Chen**, Etienne Vouga. SGP, 2025.
- *Half-Space Power Diagrams and Discrete Surface Offsets.*
Zhen Chen, Daniele Panozzo, Jeremie Dumas. IEEE TVCG, 2019.

Physics-Based Simulation & Graphics

- *Better Bending: Analysis, Construction and Verification of Discrete Bending Models for Kirchhoff-Love Shells.*
Zhen Chen, Etienne Vouga, Danny M. Kaufman. ACM TOG (SIGGRAPH), 2026. (*Equally contributed*)
- *Complex Wrinkle Field Evolution.*
Zhen Chen, Danny M. Kaufman, Mélina Skouras, Etienne Vouga. ACM TOG (SIGGRAPH), 2023.
- *Fine Wrinkling on Coarsely-Meshed Thin Shells.*
Zhen Chen, Hsiao-yu Chen, Danny M. Kaufman, Mélina Skouras, Etienne Vouga. ACM TOG (SIGGRAPH), 2021.

Patents

- Low-poly mesh generation for 3D models (US 12,333,653)
- ISO-surface mesh generation for 3D models (US 12,505,621)

Technical Expertise

- Geometry processing and mesh algorithms
- Physically-based simulation and numerical optimization
- Real-time and high-performance system design
- Simulation for interaction modeling and controllable systems

Research Interests: simulation systems, interactive graphics, generative modeling, controllable generation, and applications to embodied AI

Professional Service

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|---------------------------|-------------------|
| • SIGGRAPH | 2023–2026 |
| • SIGGRAPH Asia | 2024, 2025 |
| • Eurographics | 2024 |
| • Computer Graphics Forum | 2022 |
| • IEEE TVCG | 2023, 2025 |
| • NeurIPS | 2024 |
| • SCA | 2024 (IPC Member) |
| • ICLR | 2025, 2026 |
| • ICML | 2025 |
| • CVPR | 2026 |
| • AISTATS | 2025 |

Skills

Programming: C++, Python

Expertise: Geometry processing, simulation systems, optimization

Languages: English (fluent), Chinese (native)